Homework Solutions: Lectures 7 & 8

Confidence Intervals & Bootstrap

Probability & Statistics Course

Problem 01 · Election Night Nail-Biter

Exercise

A poll of $n = 900$ likely voters finds that 52% support candidate A.

a) Build a 95% **Wald** CI for the true proportion p. Based on this interval, can you call the election for candidate A?

b) Now compute the **Wilson** CI for the same data. How does it differ from the Wald CI?

c) Imagine only $n = 20$ voters were polled and 11 (55%) said A. Recompute both Wald and Wilson CIs. Which one behaves better near the boundary, and why?

Solution

(a) Wald CI. The Wald 95% CI for a proportion is $\hat{p} \pm z_{0.025}\,\mathrm{SE}$, where

$$\mathrm{SE}_{\mathrm{Wald}} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = \sqrt{\frac{0.52 \cdot 0.48}{900}} = 0.01665.$$

With $z_{0.025} = 1.96$:

$$\mathrm{CI} = 0.52 \pm 1.96 \cdot 0.01665 = (0.4874,\ 0.5526).$$

The lower bound $0.4874 < 0.5$, so the CI **contains** 0.5. We **cannot** call the election for candidate A at the 95% confidence level – the data is consistent with a tie or even a loss.

(b) Wilson CI. The Wilson CI is:

$$\frac{\hat{p} + \frac{z^2}{2n}}{1 + \frac{z^2}{n}} \pm \frac{z}{1 + \frac{z^2}{n}} \sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}.$$

For $n = 900$, $z^2/n = 3.8416/900 \approx 0.00427$, which is tiny. The Wilson CI evaluates to

$$(0.4873,\ 0.5525).$$

At $n = 900$ the Wald and Wilson CIs are nearly identical – the Wilson correction is negligible for large n. Both contain 0.5.

(c) Small sample: $n = 20$, $\hat{p} = 11/20 = 0.55$. **Wald:**

$$\text{SE} = \sqrt{\frac{0.55 \cdot 0.45}{20}} = 0.1112, \qquad \text{CI} = (0.3320,\ 0.7680).$$

Wilson:

$$\text{CI} = (0.3421,\ 0.7418).$$

The Wilson CI is narrower and pulled toward 0.5. The Wald CI can overshoot $[0, 1]$ for extreme $\hat{p}$ or small n because it treats the sampling distribution as symmetric. The Wilson CI accounts for the curvature of the binomial likelihood and always stays within valid bounds, making it more reliable for small samples.

Problem 02 · A/B Test: Ship It or Wait?

Exercise

An e-commerce company runs an A/B test on their checkout page.

- Control ($n_1 = 200$): conversion rate $\hat{p}_1 = 0.08$ (8%).
- Treatment ($n_2 = 200$): conversion rate $\hat{p}_2 = 0.11$ (11%).

The product manager says "3% lift – ship it!"

a) Build a 95% CI for the difference $p_2 - p_1$.

Recall: $\text{SE} = \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$.

b) Does the CI contain 0? What does this tell the PM?

c) How large would each group need to be (equal sizes) so that the expected CI width is narrow enough to exclude 0, assuming the true lift really is 3%?

Solution

(a) 95% CI for $p_2 - p_1$.

$$\hat{p}_2 - \hat{p}_1 = 0.03.$$

$$\text{SE} = \sqrt{\frac{0.08 \cdot 0.92}{200} + \frac{0.11 \cdot 0.89}{200}} = \sqrt{\frac{0.0736}{200} + \frac{0.0979}{200}} = \sqrt{0.000857} = 0.02928.$$

$$\text{CI} = 0.03 \pm 1.96 \cdot 0.02928 = (-0.0274,\ 0.0874).$$

(b) Does the CI contain 0? Yes. Since the CI includes 0, we cannot rule out that the true conversion rates are equal. The observed 3% lift is not statistically significant at the 95% level. The PM should **not** ship based on this evidence alone – either collect more data or accept the uncertainty.

(c) Required sample size. We need the margin of error to be less than the observed effect: ME < 0.03.

$$z \cdot \sqrt{\frac{p_1(1-p_1)+p_2(1-p_2)}{n}} < 0.03$$

$$n \geq \frac{z^2\big[p_1(1-p_1)+p_2(1-p_2)\big]}{0.03^2} = \frac{3.8416 \cdot (0.0736+0.0979)}{0.0009} = \frac{3.8416 \cdot 0.1715}{0.0009} \approx 733.$$

Each group needs at least $n = 733$ users – roughly $3.7\times$ the original 200.

Problem 03 · Bootstrap vs Formula

Exercise

Generate $n = 30$ observations from $\text{Exp}(\lambda = 1)$ in Python (set `np.random.seed(509)`).

a) Compute the **analytical** SE of $\bar{X}$ (since $\sigma = 1/\lambda = 1$, this is $\sigma/\sqrt{n}$).

b) Compute the **bootstrap** SE with $B = 10{,}000$ resamples.

c) Build three 95% CIs for $\mu = 1/\lambda$:

   (i) Normal-theory: $\bar{X} \pm z_{0.025} \cdot \text{SE}$

   (ii) Bootstrap percentile: $[\hat{\theta}^*_{0.025},\ \hat{\theta}^*_{0.975}]$

   (iii) t-interval: $\bar{X} \pm t_{n-1,0.025} \cdot \frac{s}{\sqrt{n}}$

   Compare widths. Do all three contain the true $\mu = 1$?

d) Repeat (a)-(c) for the **sample median** instead of the mean. Which method **cannot** give you a formula-based SE?

Solution

See the accompanying Jupyter notebook (`hw_07_08_solutions.ipynb`) for runnable code and plots.

With seed 509: $\bar{X} = 0.8061$, $s = 0.8601$, median $= 0.5374$.

(a) Analytical SE. For $\text{Exp}(\lambda = 1)$, $\sigma = 1/\lambda = 1$, so

$$\text{SE}(\bar{X}) = \frac{\sigma}{\sqrt{n}} = \frac{1}{\sqrt{30}} = 0.1826.$$

(b) Bootstrap SE. $\text{SE}_{\text{boot}} = 0.1531$, close to the analytical value. The bootstrap SE uses the sample's actual distribution rather than the known σ, so it reflects the randomness in this particular sample.

(c) Three 95% CIs for $\mu = 1$.

Method	95% CI	Width
Normal-theory (z)	(0.4483, 1.1640)	0.7157
Bootstrap percentile	(0.5288, 1.1249)	0.5961
t-interval	(0.4850, 1.1273)	0.6424

All three contain $\mu = 1$. The normal-theory interval is widest because it uses the theoretical $\sigma = 1$, which overestimates the SE for this particular sample (where $s = 0.86$). The t-interval uses s but inflates via $t_{29} = 2.045 > 1.96$. The bootstrap percentile is narrowest as it adapts to the sample's actual shape.

(d) Sample median. There is **no closed-form formula** for the SE of the median of an exponential distribution (in general, the median's SE depends on the density at the population median, which requires knowing the distribution). The bootstrap is the only practical option here.

Bootstrap SE of the median: 0.1968. Bootstrap percentile CI: (0.2412, 0.9499).

The true median of Exp(1) is $\ln 2 \approx 0.6931$, which falls inside this interval.

The normal-theory and t-interval methods **cannot** be directly applied to the median because they require a formula for the standard error. This is precisely the scenario where the bootstrap shines.

Problem 04 · Bootstrap the Correlation

Exercise

Given these (x, y) pairs:

$$(1,2),\ (2,3),\ (3,5),\ (4,4),\ (5,7),\ (6,8),\ (7,6),\ (8,9),\ (9,10),\ (10,12).$$

a) Compute the Pearson correlation r.

b) Use the bootstrap ($B = 5{,}000$) to build a 95% percentile CI for the population correlation ρ.

c) Plot the bootstrap distribution of r^*. Is it symmetric? If not, why might that be?

d) Fisher's z-transform gives an analytical CI: transform $z = \frac{1}{2}\ln\frac{1+r}{1-r}$, build a normal CI using $\mathrm{SE}(z) = \frac{1}{\sqrt{n-3}}$, then back-transform with $r = \frac{e^{2z}-1}{e^{2z}+1}$. Compare with your bootstrap CI.

Solution

See the notebook for code and histogram of the bootstrap distribution.

(a) Pearson r. $r = 0.9506$ – a strong positive linear relationship.

(b) Bootstrap 95% percentile CI. Bootstrap 95% CI: (0.8250, 0.9928).

(c) Symmetry of bootstrap distribution. **No.** The distribution is strongly left-skewed (skewness ≈ -5.0). This happens because r is bounded by $[-1, 1]$. When the true correlation is high ($\rho \approx 0.95$), there is very little room to go higher but plenty of room to go lower, creating a long left tail. This "ceiling effect" is exactly why Fisher's z-transform exists.

(d) Fisher z-transform CI.

$$z = \tfrac{1}{2}\ln\frac{1+r}{1-r} = \tfrac{1}{2}\ln\frac{1.9506}{0.0494} = 1.8384.$$

$$\mathrm{SE}(z) = \frac{1}{\sqrt{n-3}} = \frac{1}{\sqrt{7}} = 0.3780.$$

$$z \in (1.8384 - 1.96\cdot 0.3780,\ 1.8384 + 1.96\cdot 0.3780) = (1.0976,\ 2.5792).$$

Back-transform: $r = \frac{e^{2z}-1}{e^{2z}+1}$:

$$\rho \in (0.7996,\ 0.9886).$$

The Fisher CI is wider on the left (lower bound 0.80 vs bootstrap's 0.83), reflecting the fact that the z-transform properly symmetrizes the distribution. Both methods agree that ρ is strongly positive. The Fisher CI is slightly more conservative – appropriate given the small sample size ($n = 10$).

Problem 05 · When Bootstrap Breaks

Exercise

Consider $X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \mathrm{Uniform}(0, \theta)$. The MLE is $\hat{\theta} = X_{(n)} = \max(X_i)$.

a) Show that $n(\theta - X_{(n)}) \xrightarrow{d} \mathrm{Exp}(1/\theta)$.

   Hint: start from $P(X_{(n)} \le x) = (x/\theta)^n$ and substitute $u = n(\theta - x)$.

b) Using $n = 50$ and $\theta = 1$, simulate 10,000 samples. For each sample, also run $B = 1{,}000$ bootstrap resamples of $\hat{\theta}^*$. Compare the true distribution of $n(\theta - \hat{\theta})$ with the bootstrap distribution of $n(\hat{\theta} - \hat{\theta}^*)$. Do they match?

c) Explain why the bootstrap fails here. What is special about the convergence rate of $\hat{\theta}$?

   Hint: the bootstrap "works" when the convergence rate is $\sqrt{n}$. Here it is n.

Solution

(a) Limiting distribution of $n(\theta - X_{(n)})$. The CDF of the maximum is:

$$P(X_{(n)} \le x) = \left(\frac{x}{\theta}\right)^n, \qquad 0 \le x \le \theta.$$

Let $U = n(\theta - X_{(n)})$, so $X_{(n)} = \theta - U/n$. Then for $u \ge 0$:

$$P(U \le u) = P\big(X_{(n)} \ge \theta - u/n\big) = 1 - P\big(X_{(n)} < \theta - u/n\big) = 1 - \left(\frac{\theta - u/n}{\theta}\right)^n = 1 - \left(1 - \frac{u}{n\theta}\right)^n.$$

Taking $n \to \infty$:

$$\left(1 - \frac{u}{n\theta}\right)^n \to e^{-u/\theta},$$

so

$$P(U \le u) \to 1 - e^{-u/\theta},$$

which is the CDF of $\mathrm{Exp}(\text{rate} = 1/\theta)$, i.e., Exp with mean θ.

(b) Simulation. See the notebook for code and overlay histograms. The two distributions look very different:

- The **true distribution** of $n(\theta - \hat{\theta})$ is approximately Exp(1) – a smooth exponential curve on $[0, \infty)$.
- The **bootstrap distribution** of $n(\hat{\theta} - \hat{\theta}^*)$ has a point mass at 0 (when the bootstrap resample includes the original maximum) and decays differently. It does **not** match the true distribution.

(c) Why does the bootstrap fail? The bootstrap works by approximating the true sampling distribution through resampling. Its validity relies on the **convergence rate being** $\sqrt{n}$, so that the plug-in distribution and the true distribution are close in a precise sense (the bootstrap is "consistent").

For $\hat{\theta} = X_{(n)}$, the convergence rate is n, not $\sqrt{n}$. This is a **super-efficient** estimator – it converges faster than the usual parametric rate.

The key problem: the bootstrap resamples from $\{X_1, \ldots, X_n\}$, so $\hat{\theta}^* = \max(X_i^*) \leq \hat{\theta}$ **always**. This means $\hat{\theta}^* = \hat{\theta}$ with probability $1 - (1 - 1/n)^n \approx 1 - 1/e \approx 0.632$. The bootstrap distribution has a large point mass at $\hat{\theta}$, while the true distribution of $\hat{\theta}$ has no point mass at θ. The bootstrap fundamentally cannot reproduce the behavior near the boundary, so it gives invalid confidence intervals.

Remedy: For boundary estimators like this, use parametric bootstrap (simulate from Uniform$(0, \hat{\theta})$) or exact methods based on the known limiting distribution Exp$(1/\theta)$.